

EX NO: 1b	CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB
DATE:	

PROGRAM (MATLAB CODE)

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clc;

clear;

close all

Am = 1;

fm = 500;

Ac = 5;

fc = 5000;

fs = 20*fc;

t = 0:1/fs:4/fm;

% Message and Carrier Signals

m = Am*sin(2*pi*fm*t);

c = Ac*sin(2*pi*fc*t);

% Modulation Index

mu = Am/Ac;

% AM Signal

am = Ac*(1 + mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);

% Demodulation using Hilbert Transform

dem = abs(hilbert(am));

dem = dem - mean(dem);

% FFT Analysis

N = length(am);

f = linspace(-fs/2, fs/2, N);

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AM_FFT = abs(fftshift(fft(am)))/N;

% Sideband Frequencies

USB = fc + fm;

LSB = fc - fm;

% Power Calculation

Pc = Ac^2/2;

Psb = (mu^2 * Pc)/4;

% Plotting

figure('Name','AM Modulation','NumberTitle','off');

subplot(3,2,1);

plot(t,m);

title('Message Signal');

grid on;

subplot(3,2,2);

plot(t,c);

title('Carrier Signal');

grid on;

subplot(3,2,3);

plot(t,am);

title('AM Signal');

grid on;

subplot(3,2,4);

plot(t,dem);

title('Demodulated Signal');

grid on;

subplot(3,2,5);

plot(f,AM_FFT);

xlim([fc-3*fm fc+3*fm]);

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title('Frequency Spectrum (Sidebands)');

xlabel('Frequency (Hz)');

ylabel('Magnitude');

grid on;

subplot(3,2,6);

bar([Pc Psb Psb]);

set(gca,'XTickLabel',{'Carrier','USB','LSB'});

title('Power Distribution');

ylabel('Power (W)');

grid on;

% Display Results

fprintf('Modulation Index ( $\mu$ ) = %.2f\n', mu);

fprintf('Upper Sideband (USB) = %d Hz\n', USB);

fprintf('Lower Sideband (LSB) = %d Hz\n', LSB);

fprintf('Carrier Power ( $P_c$ ) = %.2f W\n', Pc);

fprintf('Sideband Power (Each) = %.3f W\n', Psb);

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OUT PUT

